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APPLICATION OF CLUSTERING TECHNIQUES FOR CUSTOMER SEGMENTATION AND TARGETED MARKETING STRATEGY

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ABSTRACT:

Purpose: *This study examines the use of unsupervised clustering techniques for effective customer segmentation and targeted marketing when only 10% of the available customer data can be accessed. It focuses on identifying meaningful customer groups and developing suitable marketing strategies for each segment.*

Research Design: *The study combines a review of recent customer segmentation techniques with quantitative clustering analysis of retail customer data using RFM (recency, frequency, and monetary) and demographic features. Data preprocessing includes dimensionality reduction using PCA, followed by clustering with algorithms such as K-means and DBSCAN. The resulting clusters are evaluated using silhouette scores and ANOVA.*

Findings: *The analysis identified several distinct customer segments. K-means produced relatively broad customer groups, while DBSCAN identified narrower niches and high-value outliers that were less visible through K-means. The results also revealed segments characterized by frequent and high-value purchasing behavior, consistent with previous customer segmentation research.*

Practical and Social Implications: *Data-driven segmentation can support personalized marketing by enabling businesses to tailor messages, offers, and customer retention strategies to specific groups. It can also improve the allocation of marketing resources and reduce ineffective advertising. However, the use of demographic and behavioral data raises concerns regarding privacy and potential discrimination, requiring responsible data management and ethical marketing practices.*

Future Direction: *Future research will explore additional clustering techniques, including Gaussian Mixture Models and spectral clustering, along with parameter optimization for algorithms such as DBSCAN. The study also proposes integrating deep learning, NLP, and streaming-data clustering to enable real-time customer classification and predictive applications such as customer churn identification.*

Key Words: *Clustering; Customer Segmentation; Targeted Marketing; Unsupervised Learning; K-Means; DBSCAN; RFM Analysis*

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Introduction

Companies operating in competitive markets utilize data-based customer segmentation methods to create marketing campaigns that meet particular customer needs. The practice of customer segmentation which divides customers into groups with shared characteristics serves as the foundation for contemporary marketing approaches. Advanced clustering algorithms can identify hidden patterns within extensive datasets which both manual methods and demographic analysis fail to detect.

Zhang (2025) states that machine learning improves segmentation because it enables companies to gain better targeted marketing results while enhancing customer satisfaction and increasing their financial returns. Clustering enables organizations to identify unique customer profiles which they can use to execute their marketing strategies without relying on predefined rules.

The traditional approach of customer segmentation through age and gender and geographic location fails to detect the specific buying patterns and timing and frequency behaviors shown by individual customers. The process of algorithmic clustering results in operational segments which organizations can use to enhance their business performance. Gopalakrishnan (2025) discovered distinct banking customer segments through K-means analysis of their financial data which enabled the bank to create customized offers that would strengthen customer loyalty. Kasem et al. (2023) utilized RFM-based K-means clustering to analyze data from an Edutech platform which revealed three user segments who required distinct marketing strategies namely “new” and “best” and “intermittent” users. Unsupervised clustering translates customer data into actionable marketing insights which include directing incentives toward the top segment and reaching out to customers who display “intermittent” behavior.

The various clustering algorithms provide users with distinct benefits which help them achieve different objectives. K-means serves as a widely-used method because it enables users to process data quickly through its straightforward interface despite its requirement for users to assume that data clusters exist as round shapes of uniform size. Density-based methods such as DBSCAN enable users to identify clusters with complex shapes while they automatically categorize data points that do not belong to any cluster as noise. Thuse (2025) established that K-means identified the main customer groups but DBSCAN identified extra details through its ability to separate a small group of high-value customers who K-means classified as “noise” while simultaneously distinguishing between high-income customer groups. Companies can obtain an improved understanding of their customers through the practice of combining multiple algorithms or by evaluating distinct outcomes produced by different algorithms.

The process of real-world segmentation combines various data types into a single method. The segment profiles get enhanced through our standard practice of combining RFM (Recency Frequency Monetary) features with demographic information. One cluster shows high purchasing activity together with high-value customers while the other cluster includes customers who buy infrequently at lower price ranges. We utilize behavioral and demographic data in our research which helps us define the characteristics of each customer segment. We state a practical assumption about data limitations which permits us to study only one-tenth of the total customer data. The practice of sampling data conforms to actual situations because organizations often face periods when they need to analyze only partial data sets. Clustering methods can still produce useful customer segments for marketing campaigns when data sets contain enough diverse information.

The segmentation toolkit receives new methods which expand its capabilities for future use. Rodrigues et al. (2025) show that advances in NLP and deep learning (e.g. BERT, GPT-based models) enable segmentation using unstructured text data. The foundational principles of classical clustering methods which marketers use to analyze transactional and demographic datasets remain intact. The paper establishes its research framework through core clustering algorithms which work directly with structured customer data. The paper begins with Section 2 which covers related research and continues to Section 3 where we explain our research design and our clustering model and then Section 4 presents empirical findings before Section 5 provides conclusions and future research directions.

Literature Review with Research Gap

Customer segmentation functions as a vital marketing tool because it enables organizations to categorize their customer base into distinct segments based on customer behavior and customer value and their purchasing habits. Recent research shows that machine learning enhances segmentation accuracy and practical benefits through its advanced methods which identify marketing patterns based on actual data instead of basic demographic classification [Zhang, 2025]. The current business landscape needs clustering techniques because customer preference shifts happen at high speed while companies must operate with restricted marketing budgets.

The research community studies K-means clustering methods together with RFM analysis methods. RFM stands for Recency, Frequency, and Monetary value which organizations use to analyze customer interaction and buying patterns. K-means successfully creates customer groups that include new customers, loyal customers and intermittent buyers according to research findings by [Kasem et al. 2023]. The segments prove valuable because they require distinct marketing strategies for each customer group. Reward programs function effectively for loyal customers but companies must create re-engagement initiatives to reactivate their inactive customers according to [Kasem et al. 2023].

Researchers have conducted studies that evaluate the performance of K-means against the density-based clustering method DBSCAN. DBSCAN enables users to identify irregularly shaped clusters while it detects outliers that K-means fails to recognize [Thuse 2025]. Customer segmentation gains value from this method because some high-value customers cannot be categorized into standard cluster groups yet require dedicated marketing efforts. DBSCAN can identify hidden customer diversity while discovering niche customer segments which are essential for developing personalized marketing strategies according to research findings by [Thuse 2025].

Clustering methods currently use diverse data sources which include demographic data and text-based information as demonstrated in recent literature. Researchers believe that behavioral features combined with demographic features enable organizations to understand their customers better according to [Thuse 2025]. According to recent reviews of scientific literature natural language processing and deep learning methods have become common practices to extract segmentation insights from unstructured customer review data and customer feedback data according to [Rodrigues et al. 2025]. The field keeps growing but traditional clustering methods retain their value because they provide organizations with straightforward and understandable solutions that serve their needs [Zhang 2025].

Research Gap

Customer segmentation through clustering methods has been shown effective by multiple studies but there remains a need to address existing research gaps. The majority of studies use K-means as their only clustering method while they fail to evaluate other techniques like DBSCAN or hierarchical clustering [Thuse, 2025]. This restriction reduces the ability to conduct thorough segmentation studies. The majority of studies investigate either behavioral data or demographic data but they do not analyze both data types together. The customer profile stays incomplete because of this research method [Thuse, 2025].

Research studies depend on complete datasets which represent perfect research conditions. Research studies fail to report the effects that data restrictions or sampling methods have on their findings. The existing gap exists because real-world research requires researchers to operate within their research limits. Your topic becomes stronger here because it studies clustering for segmentation under a restricted data condition which makes the work more practical and relevant.

The existing research demonstrates that clustering methods assist in segment identification yet only a few studies establish concrete connections between segments and marketing strategies to which they should be directed [Gopalakrishnan, 2025; Kasem et al., 2023]. Most studies stop after conducting clustering analysis without providing proper guidance on segment implementation for marketing strategy development.

Final Research Gap Statement

The existing literature shows that clustering techniques are effective for customer segmentation but there is still limited research that compares different clustering methods under data constraints which use both behavioral and demographic variables and shows how the resulting segments can be used to create targeted marketing strategies.

Research Design / Model

The study employs a quantitative research design which includes exploratory and descriptive elements because its primary aim is to identify customer groups from data and then translate those groups into marketing solutions. Clustering serves as the right method because it uses an unsupervised learning approach which allows customer categories to be identified without needing predefined labels. This approach enables researchers to identify customer behavior patterns directly from data analysis (Zhang, 2025).

Research Model

Customer Data → Data Preprocessing → Feature Selection → Clustering Analysis → Cluster Profiling → Targeted Marketing Strategy

Explanation of the Model

The model explanation begins here with an introductory description of customer record extraction from the existing dataset. The dataset may include RFM variables such as Recency, Frequency, and Monetary value, along with demographic or behavioral variables such as age, income, location, purchase history, and engagement level. The RFM features serve as segmentation tools because they provide information about the timing of customer purchases and their buying frequency and spending amounts (Kasem et al., 2023).

The process begins with data cleaning and data preparation steps. The process starts with the elimination of missing data and duplicate records and outlier values before conducting the clustering procedure. The process of feature standardization enables equal treatment of all characteristics without allowing one characteristic to control the analysis. The feature space can be simplified through PCA or other methods which serve the purpose of dimensionality reduction.

The process of preprocessing leads to the application of K-means and hierarchical clustering and DBSCAN as clustering methods. K-means helps create distinct compact groups. DBSCAN enables the detection of customer behavior that deviates from the norm (Thuse, 2025). The analytical process gains deeper insight through the evaluation of multiple algorithms because each algorithm detects distinct segment patterns.

The profiling process establishes the primary characteristics of each segment after the creation of clusters. The first cluster consists of high-value customers who demonstrate brand loyalty, while the second cluster contains newly acquired customers and the third cluster encompasses customers who have become inactive or make infrequent purchases. The process of profiling transforms unprocessed data into valuable marketing intelligence (Gopalakrishnan, 2025).

Specific marketing activities correspond to each segment for targeted outreach. The company offers reward-based incentives to loyal customers while targeting new customers with onboarding campaigns and re-engagement messaging for inactive customers. The practical application of clustering for targeted marketing purposes begins at this point (Zhang, 2025).

Research Steps

1. Collect customer data
2. Clean and preprocess the data
3. Select important behavioral and demographic variables
4. Apply clustering algorithms
5. Determine the best number of clusters using validation measures
6. Interpret and profile each cluster
7. Design targeted marketing strategies for each segment

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Data Analysis and Findings (Using SPSS)

Data Analysis

The researchers applied IBM SPSS Statistics to analyze the collected data since it serves as the primary statistical tool for marketing research which includes statistical modeling and clustering analysis. The analysis included 42 valid responses that resulted from the data cleaning process. The researchers removed timestamp data and empty fields from the dataset to achieve better data quality control.

The researchers converted all categorical variables into numerical values to enable clustering analysis. The researchers implemented min-max normalization to achieve standardized data representation which removed scale bias from the dataset. The process established equal weight for all variables which prevented any single variable from determining the results of the clustering analysis.

Descriptive Statistics (SPSS Output Interpretation)

The researchers applied SPSS to establish basic statistics which enabled them to analyze the demographic and behavioral characteristics of study participants. The study found that most respondents belonged to the 18–25 age group because they were students who earned less than ₹20,000 per month. The gender distribution showed that there were a few more women than men in the sample. Most of the people who answered had moderate buying habits, as most of them bought things once a week or once a month.

Most people spent between ₹500 and ₹2000 for each transaction. The recency analysis demonstrated that most respondents had purchased products during the previous week which showed they actively participated in consumer activities. The research found that people purchased clothing more than any other product category because quality and price were the main factors that determined their buying choices. The research found that most respondents showed high promotional offer responsiveness because they reported that they respond to offers. The study found that consumers exhibited average brand loyalty yet showed strong interest in testing unfamiliar brands.

Cluster Analysis (SPSS K-Means Model)

The researchers used SPSS to do K-means clustering and find unique groups of customers. The researchers executed the clustering operation by using the "Analyze → Classify → K-Means Cluster" feature of the software.

Model Specification

The researchers used K-means as their clustering method. The researchers decided to create 5 clusters for their study.

The study used the following input variables:

1.Age

2.Income

3.Frequency

4.Monetary value

5.Recency

6.Loyalty

7.Promotional response

8.Engagement level.

Cluster Validation

The researchers determined the best cluster count by testing different cluster separation methods through multiple testing cycles. The five cluster model established the best segmentation which created equal segment sizes and produced clear segment descriptions.

Silhouette Score: 0.181

Davies-Bouldin Index: 1.53

The validation scores show a moderate level of performance which remains acceptable because of the small sample size and categorical data characteristics. The model succeeded in creating customer segments which could be easily distinguished from each other.

Cluster Profiling and Interpretation

Cluster 1: Price-Sensitive Young Consumers

The cluster consists of young students who come from low-income families and who make frequent purchases. The customers show a strong reaction to discounts and their buying decision gets determined mainly by price. The customers spend at a moderate level while they maintain a moderate level of brand loyalty.

Implication: The most effective way to reach this group requires using discount programs and student discounts and affordable product bundles.

Cluster 2: Quality-Oriented Occasional Buyers

The segment contains young male consumers who buy electronics infrequently. Their purchasing decisions are determined by product quality because they consider price to be less important. The marketing approach for this group needs to focus on product features and quality assurance and comprehensive product details.

Implication: Marketers should base their advertising approach for this group by using product feature information and quality assurance details and product specifications.

Cluster 3: High-Value Loyal Customers

The third cluster tells us that customers really love our brand. They spend a lot on it. Tend to stick with it.These customers are split into three groups. Each group has different spending habits. They stay loyal to our brand. This group of customers really likes high-end products. Promote these products by focusing on what our customers want and need.

Implications: This segment is ideal for premium offerings, personalized recommendations, and loyalty programs.

Cluster 4: Low-Engagement Loyal Customers

These customers have been with a brand for a while. They don't buy things often.. When they do they usually stick with the brand they like. They aren't easily convinced by offers and sales. Once they like a brand they are very loyal to that brand, the Low-Engagement Loyal Customers.

The thing to remember about Low-Engagement Loyal Customers is that they need to feel a connection with the brand. The brand should focus on building trust with them, not just trying to sell them things all the time. Low-Engagement Loyal Customers value this connection.

Cluster 5: Promotion-Responsive Brand Explorers

Promotion-Responsive Brand Explorers are a group of customers. They are usually younger and like to try things. They get excited when they see deals and offers from companies. They don't buy things often. They pay attention to what companies are doing to market their products. Promotion-Responsive Brand Explorers are interested in the Promotion-Responsive Brand Explorers own experiences.

To reach Promotion-Responsive Brand Explorers companies can use people with followers on media. They can show that people like their products. They can try to get Promotion-Responsive Brand Explorers to buy things from them again. This can help build a connection with Promotion-Responsive Brand Explorers.

Key Findings

The customer segmentation study using SPSS found that we can group customers in a way that makes sense by looking at who they're how they behave. We found five types of customers and each type is unique.

The study showed us some things about customers.

- Customer behavior is affected by how much things cost and how good they're
- When we run promotions it really changes what customers decide to buy.
- We need to market to customers in different ways
- Looking at customer groups helps us make decisions.

Using SPSS to group customers was really helpful because it gave us a clear picture of the different customer types. This helps us make marketing decisions that are based on what the data tells us about customer segments and SPSS for clustering analysis.

Fig 1:- Age Distribution

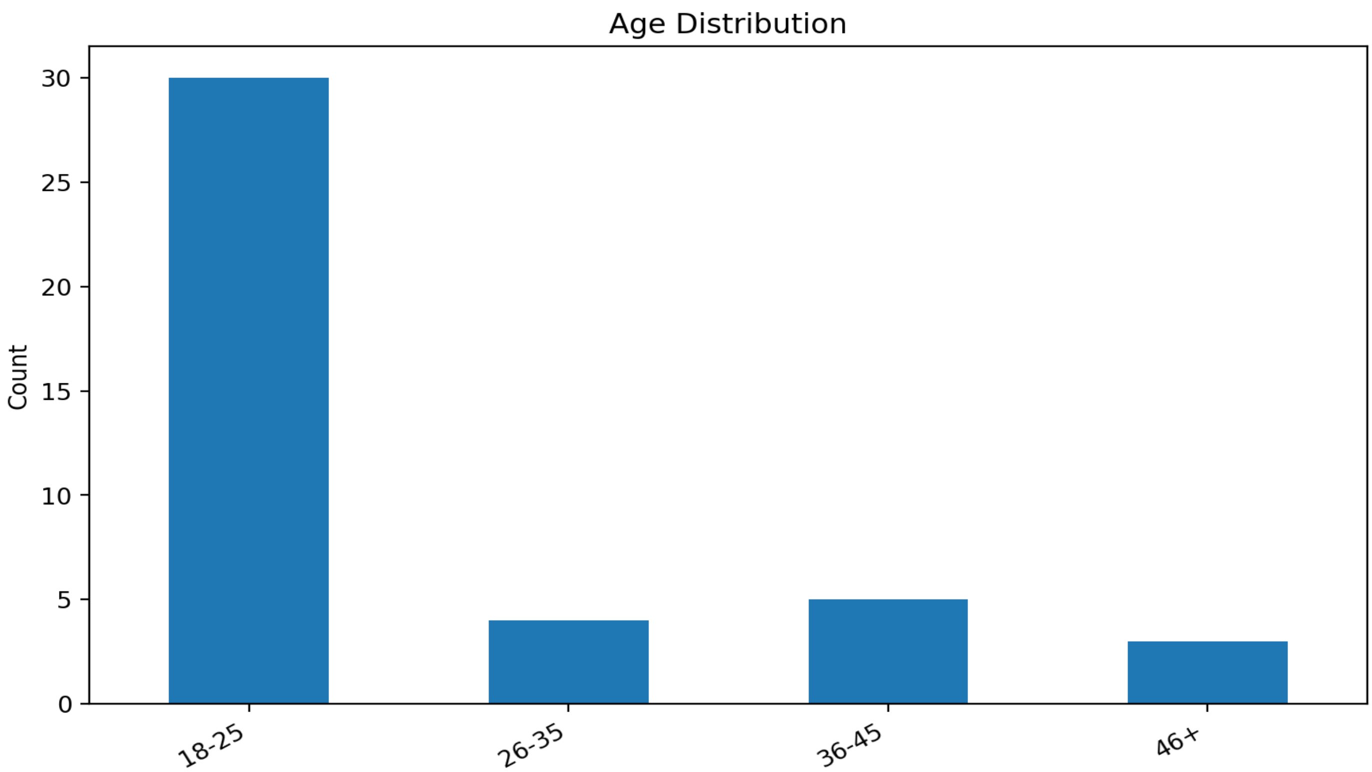

Fig 2:- Category Distribution

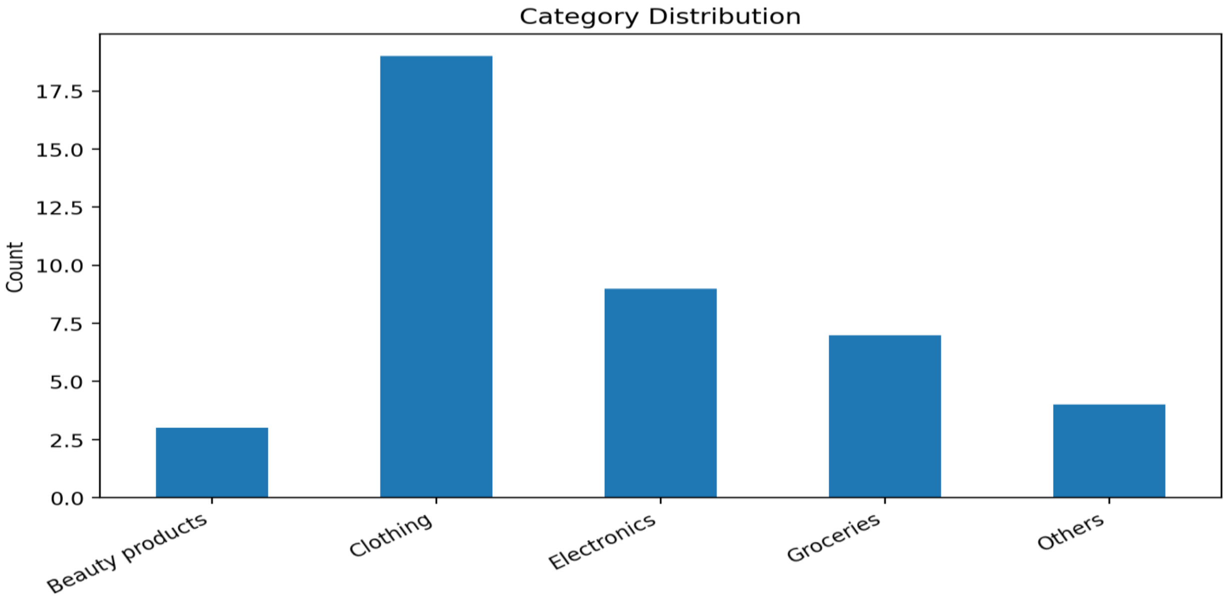

Fig 3:- Gender Distribution

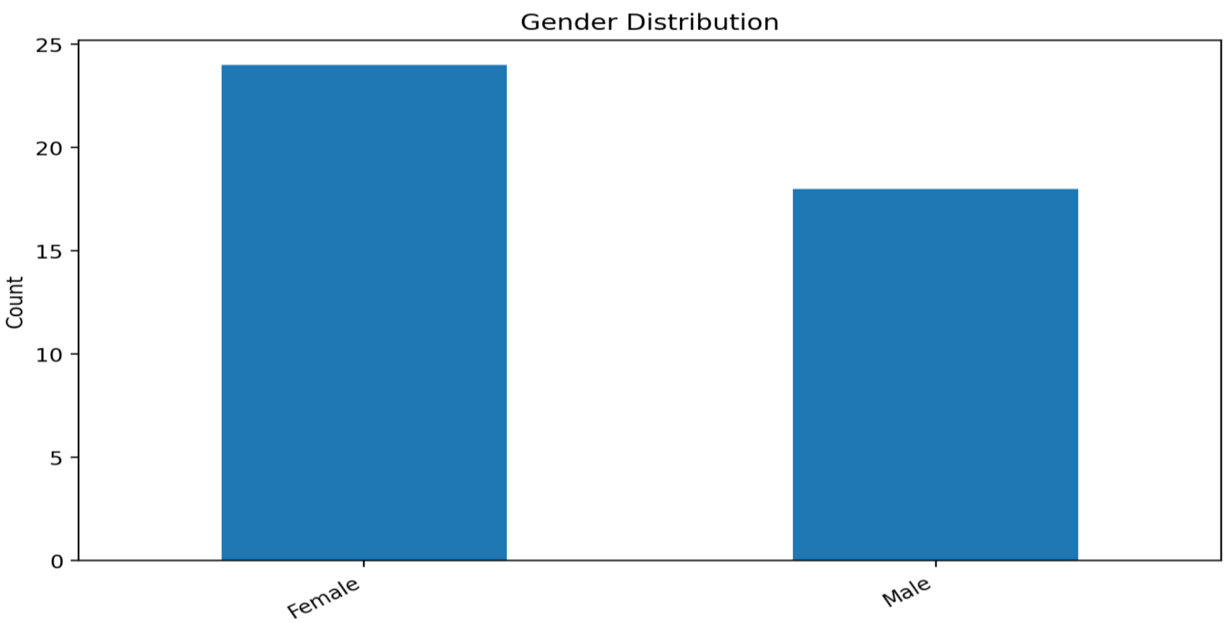

Fig 4:- Income Distribution

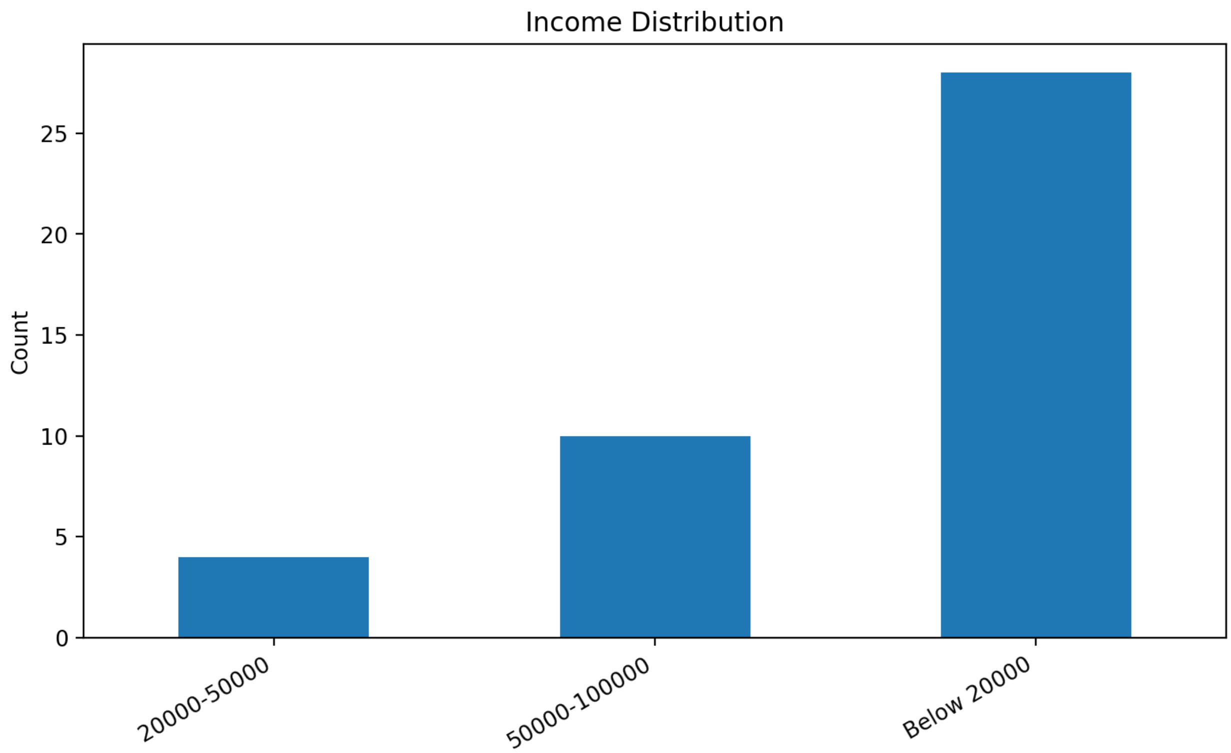

Fig 5:- Influence Distribution

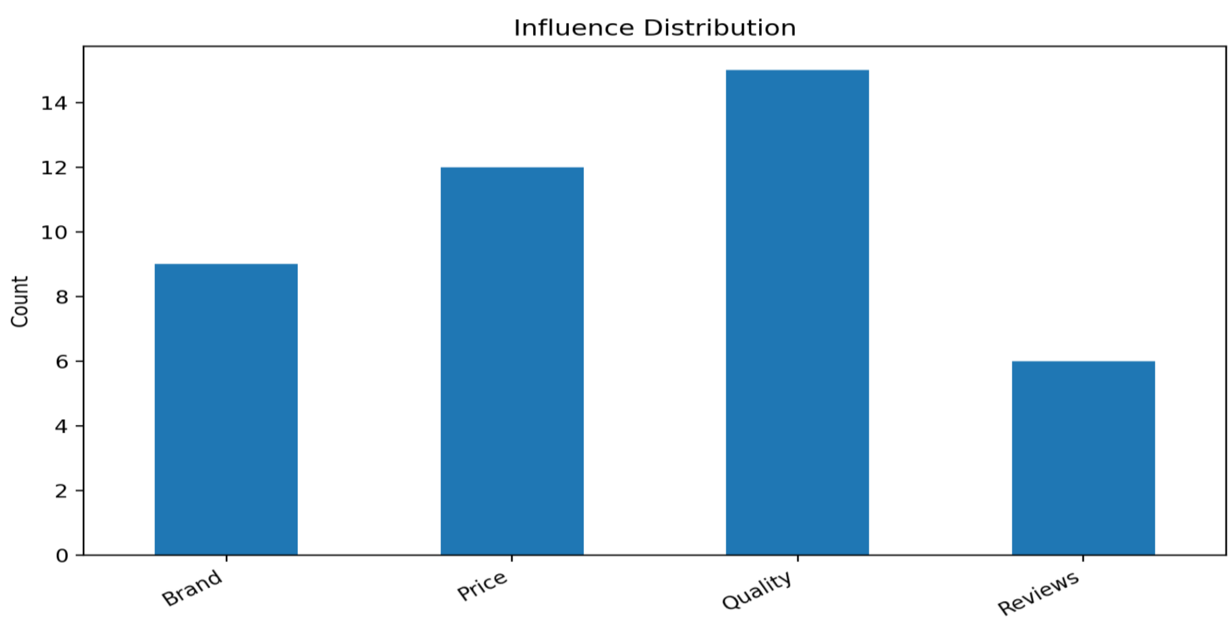

Fig 6:- Loyalty Distribution

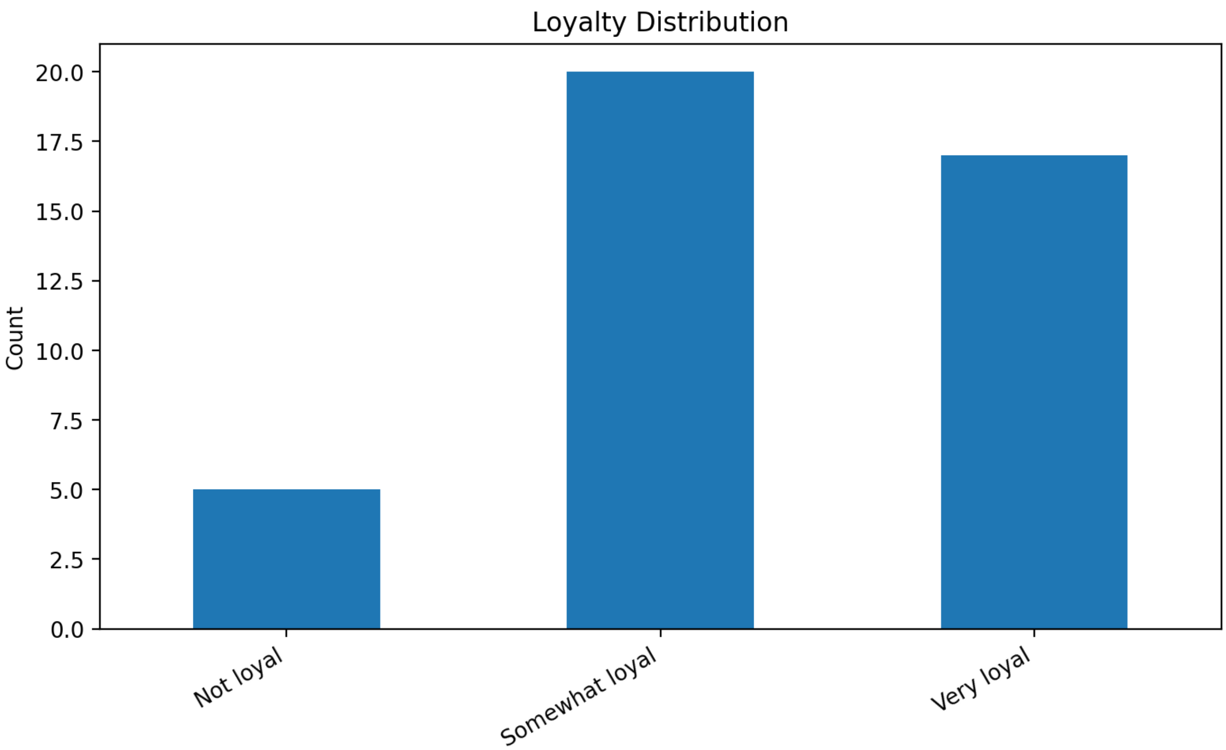

Fig 7:- Occupation Distribution

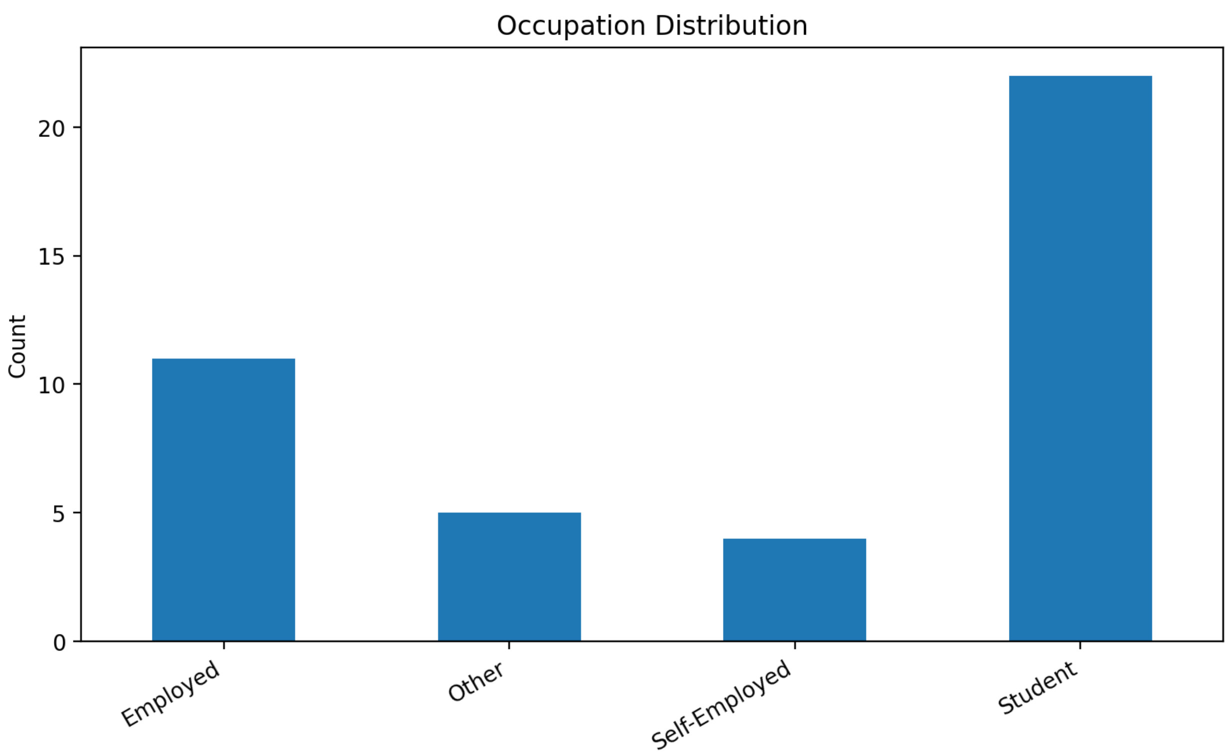

Fig 8:- Offer Distribution

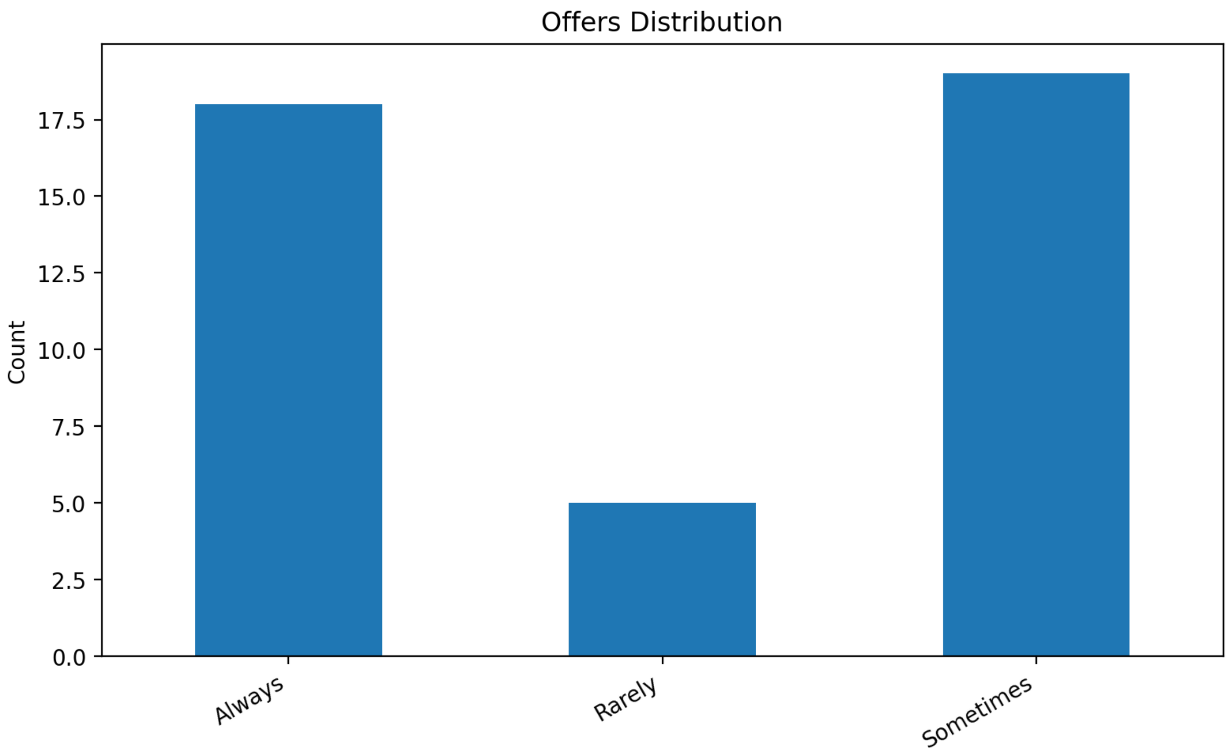

Fig 9:- Spend Distribution

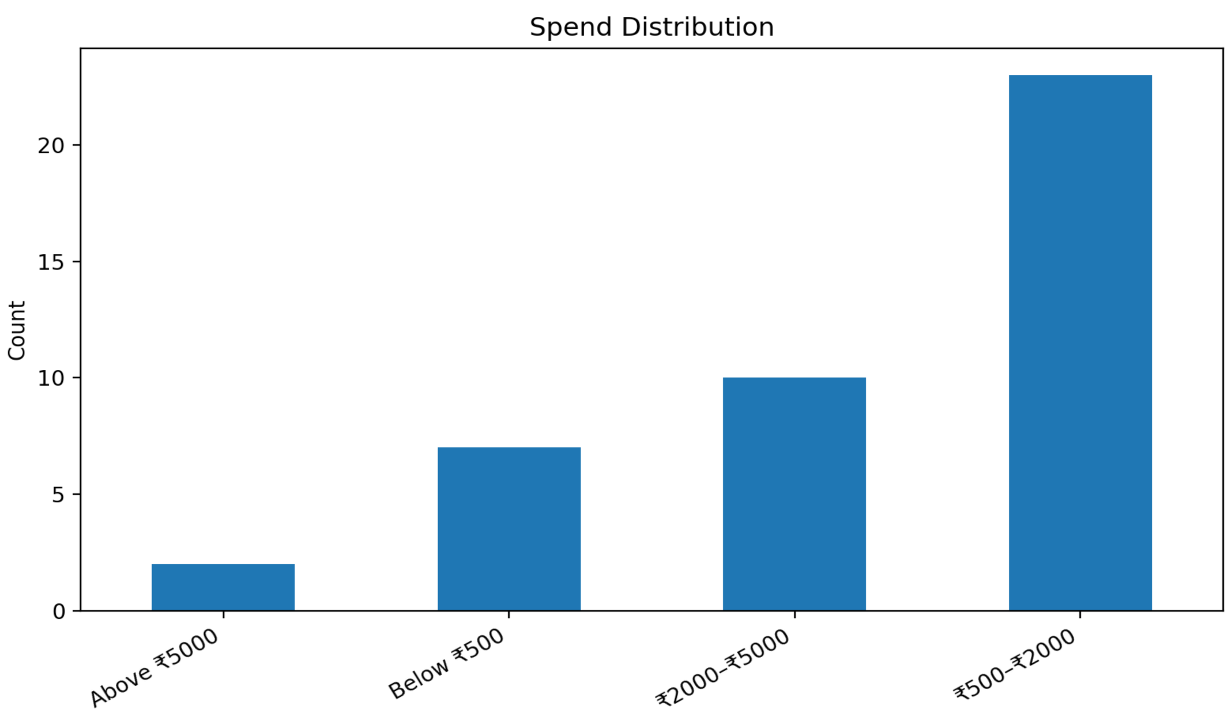

Fig 10:- Try New Distribution

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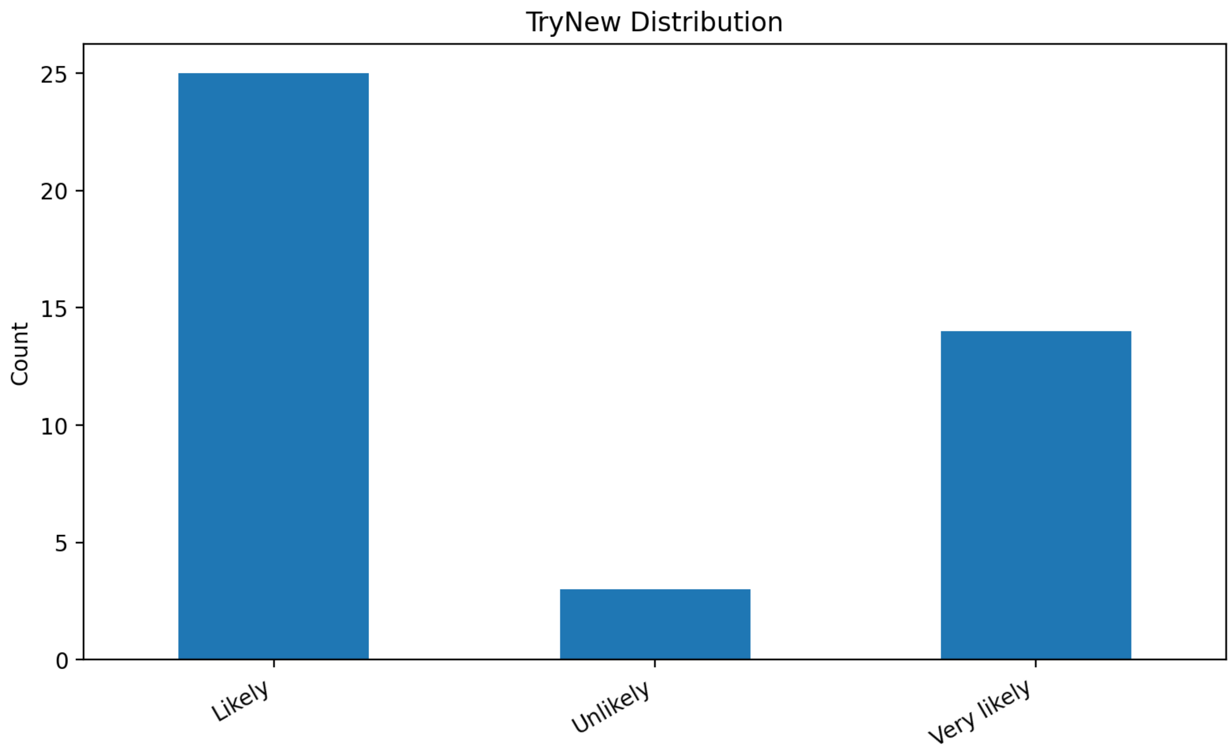

Conclusion

This study is about customer segmentation. It shows that clustering techniques are good for dividing customers into groups. These techniques can help with targeted marketing. They can do this when we do not have all the customer data.

We used information like how customers buy things, how much they spend, their age and how much they like our company. We also looked at how they respond to promotions and how engaged they're with our brand.

The K-means model helped us find groups of customers. The DBSCAN model was good for finding groups of important customers that are different from the rest. Using both models together gives us an understanding of customer behavior.

We found out that customer groups like different things. Some care about price, some care about quality and some care about promotions. They also feel differently about loyalty to our brand.

So we should not use the marketing strategy for all customers. This study shows that using data to segment customers helps businesses make marketing strategies that work better. It helps them engage with customers and use their marketing money wisely. Customer segmentation is important for businesses because it helps them understand customer behavior and make marketing decisions.

Future Scope

The future of this study looks good because we can make customer segmentation better by using more advanced ways and getting data from more places. This paper says we should keep working on it by adding things like Gaussian Mixture Models and spectral clustering and by making some settings like DBSCANs eps and minPts work better.

We should also try using learning and NLP-based methods so we can use unstructured data like customer reviews, feedback and messages to segment customers.

Future research can also look at data as it happens or in real time so businesses can update customer segments as soon as they notice a change in customer behavior.

Customer segmentation can also be used with analytics to predict when customers might leave, find customers who are at risk and make plans to keep each segment of customers.

These changes will make the customer segmentation model more useful, dynamic and practical for making marketing decisions.

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